

ELTE IK - Computer Science BSc

Final Exam Topics

2. Differential and integral calculus

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Differential and integral calculus

Differentiability of functions. Partial derivative, total derivative. Gradient, Jacobian matrix. Function analysis, extrema. Riemann integral, integration by parts, integration by substitution. Newton–Leibniz formula.

1 Differentiability of functions

Differentiability

$$1 \leq n, m \in \mathbb{N}, \quad 1 \leq p, q \leq +\infty,$$

$(\mathbb{R}^n, \|\cdot\|_p)$ and $(\mathbb{R}^m, \|\cdot\|_q)$ normed spaces

$$f \in \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad a \in \text{int}D_f$$

The function f is differentiable at the point a ($f \in D\{a\}$), if

there exists a bounded linear map $L \in L(\mathbb{R}^n, \mathbb{R}^m)$ and a function $\eta \in \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that:

$$f(a+h) - f(a) = L(h) + \eta(h) \cdot \|h\|_p \quad (h \in \mathbb{R}^n, a+h \in D_f)$$

where

$$\eta(h) \longrightarrow 0 \quad (\|h\|_p \rightarrow 0)$$

In other words:

$$\frac{f(a+h) - f(a) - L(h)}{\|h\|_p} \longrightarrow 0 \quad (\|h\|_p \rightarrow 0)$$

If $\forall a \in \text{int}D_f : f \in D\{a\}$, then f is differentiable ($f \in D$)

Remark:

A continuous map between the normed spaces $(X, \|\cdot\|_\star)$, $(X, \|\cdot\|_\heartsuit)$ over the field $\mathbb{K}$ is a bounded linear map if

- it is linear, that is,

$$f(x + \lambda y) = f(x) + \lambda f(y) \quad (x, y \in X, \lambda \in \mathbb{K})$$

- it is bounded, that is,

$$\exists M \geq 0 : \|f(x)\|_{\heartsuit} \leq M\|x\|_{\star} \quad (x \in X)$$

Derivative

the function $f \in \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at a point $a \in \text{int}D_f$

$\Rightarrow \exists! L \in L(\mathbb{R}^n, \mathbb{R}^m)$ bounded linear map

This uniquely existing bounded linear map $L \in L(\mathbb{R}^n, \mathbb{R}^m)$ is called the derivative of the function f at the point a , and is denoted by the symbol $f'(a)$.

2 Partial derivative, total derivative

2.1 Partial derivative

Definition

Consider the function $h \in \mathbb{R}^n \rightarrow \mathbb{R}$ and the vector $a = (a_1, \dots, a_n) \in D_h$. Let

$$D_{h,i}^{(a)} := \{t \in \mathbb{R} : (a_1, \dots, a_{i-1}, t, a_{i+1}, \dots, a_n) \in D_h\} \quad (i = 1, \dots, n)$$

And let:

$$h_{a,i} : D_{h,i}^{(a)} \rightarrow \mathbb{R}, \quad \text{for which:} \quad h_{a,i}(t) := h(a_1, \dots, a_{i-1}, t, a_{i+1}, \dots, a_n) \quad (t \in D_{h,i}^{(a)})$$

The partial functions $h_{a,i}$ are all single-variable real functions ($h_{a,i} \in \mathbb{R} \rightarrow \mathbb{R}$)

The function h is partially differentiable at a with respect to the i -th variable if $h_{a,i} \in D\{a_i\}$. Then:

$$\partial_i h(a) := h'_{a,i}(a_i)$$

the real number is called the partial derivative of the function h at a with respect to the i -th variable.

Partial derivative function

Suppose that for the previous function h :

$$D_{h,i} := \{a \in D_h : \text{the partial derivative } \partial_i h(a) \text{ exists}\} \neq \emptyset \quad (i = 1, \dots, n)$$

Then the function

$$x \mapsto \partial_i h(x) \quad (x \in D_{h,i})$$

is called the partial derivative function of the function h with respect to the i -th variable, and is denoted by the symbol $\partial_i h$.

Differentiability and partial differentiability

- Differentiability $\Rightarrow$ partial differentiability
 $1 \leq n \in \mathbb{N}$, $h \in \mathbb{R}^n \rightarrow \mathbb{R}$, and $h \in D\{a\}$ ($a \in D_h$)
 $\Rightarrow \forall i = 1, \dots, n$: the function h is partially differentiable at the point a with respect to the i -th variable,
 and

$$\text{grad}h(a) = (\partial_1 h(a), \dots, \partial_n h(a))$$

- Differentiability $\Leftarrow$ partial differentiability
 $1 \leq n \in \mathbb{N}$, $h \in \mathbb{R}^n \rightarrow \mathbb{R}$, and $a \in \text{int}D_h$

For some $i = 1, \dots, n$:

- At any point $x \in K_r(a)$ ($r > 0$ suitable) the partial derivatives $\partial_j h(x)$ exist ($i \neq j = 1, \dots, n$) and these are continuous
- $\exists \partial_i h(a)$ partial derivative

$\Rightarrow h \in D(a)$

2.2 Total derivative

The function $f : \mathbb{R}^p \rightarrow \mathbb{R}^q$ is said to be (totally) differentiable at the point $a \in D(f)$, if there exists a linear map $L : \mathbb{R}^p \rightarrow \mathbb{R}^q$ such that

$$\lim_{x \rightarrow a} \frac{f(x) - f(a) - L(x - a)}{\|x - a\|} = 0.$$

The linear map L is called the *differential* of the function f at the point a , and the matrix M_L of the map L is called the *differential quotient or Jacobian matrix* of f at a . Notation: $L = Df(a)$, and $M_L = f'(a)$.

For real f ($q = 1$) the Jacobian matrix $f'(a)$ is of type $1 \times p$, that is, a p -dimensional row vector. In this case, instead of the Jacobian matrix at a , the name *gradient* or *gradient vector* at a and the notation $f'(a) = \text{grad } f(a)$ are also used.

3 Jacobian matrix, gradient

Jacobian matrix

For the bounded linear map $L := f'(a)$ appearing above, $\exists! A \in \mathbb{R}^{m \times n}$ matrix for which:

$$L(x) = Ax \quad (x \in \mathbb{R}^n)$$

Therefore: $f'(a) := A$

is the derivative or derivative matrix of the function f at a , also known as its Jacobian matrix.

Gradient

For $m = 1$: $f \in \mathbb{R}^n \rightarrow \mathbb{R}$

$$\text{grad } f(a) := f'(a) \in \mathbb{R}^{1 \times n} \approx \mathbb{R}^n$$

So in this case the Jacobian matrix $f'(a)$ can be regarded as a vector in $\mathbb{R}^n$, which is called the gradient of the function f at a .

If $D := \{a \in D_f : f \in D\{a\}\}$, then the function

$$x \mapsto \text{grad } f(x) \in \mathbb{R}^n \quad (x \in D)$$

is called the gradient of the function f , and is denoted $\text{grad } f \in \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Gradient as a row of the Jacobian matrix

Let $1 \leq n, m \in \mathbb{N}$. The function $f = (f_1, \dots, f_m) \in \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at the point $a \in \text{int } D_f$ if and only if for every $i = 1, \dots, m$ the coordinate function $f_i \in \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at a .

If $f \in D\{a\}$, then the Jacobian matrix $f'(a)$ has the following form:

$$f'(a) = \begin{bmatrix} \text{grad } f_1(a) \\ \text{grad } f_2(a) \\ \vdots \\ \text{grad } f_m(a) \end{bmatrix}$$

4 Extrema, function analysis

4.1 Extremum

$$f \in \mathbb{R} \rightarrow \mathbb{R}, a \in D_f$$

Local maximum

f has a local maximum at a if for a suitable $r > 0$:

$$f(x) \leq f(a) \quad (x \in D_f, |x - a| < r)$$

Local minimum

f has a local minimum at a if for a suitable $r > 0$:

$$f(x) \geq f(a) \quad (x \in D_f, |x - a| < r)$$

Absolute maximum

f has an absolute maximum at a if:

$$f(x) \leq f(a) \quad (x \in D_f)$$

Absolute minimum

f has an absolute minimum at a if:

$$f(x) \geq f(a) \quad (x \in D_f)$$

Local extremum

f has a local extremum at a if it has a local minimum or maximum at a .

Absolute extremum

f has an absolute extremum at a if it has an absolute minimum or maximum at a .

First-order necessary condition (for a local extremum)

the function $f \in \mathbb{R} \rightarrow \mathbb{R}$ has a local extremum at the point $a \in \text{int}D_f$, and $f \in D\{a\}$

$$\Rightarrow f'\{a\} = 0$$

With the help of the previous theorem it is no longer difficult to prove the so-called mean value theorems, which are of fundamental importance for the analysis of differentiable functions.

Rolle's theorem

$a, b \in \mathbb{R}$ ($a < b$), $f : [a, b] \rightarrow \mathbb{R}$, $f \in C$, and

$\forall x \in (a, b) : f \in D\{x\}$ and $f(a) = f(b)$

$$\Rightarrow \exists \xi \in (a, b) : f'(\xi) = 0$$

Lagrange mean value theorem

$a, b \in \mathbb{R}$ ($a < b$), $f : [a, b] \rightarrow \mathbb{R}$, $f \in C$, and

$\forall x \in (a, b) : f \in D\{x\}$

$$\Rightarrow \exists \xi \in (a, b) : f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

Cauchy mean value theorem

$a, b \in \mathbb{R}$ ($a < b$), $f, g : [a, b] \rightarrow \mathbb{R}$, $f, g \in C$, and

$\forall x \in (a, b) : f, g \in D\{x\}$

$$\Rightarrow \exists \xi \in (a, b) :$$

$$(f(b) - f(a)) \cdot g'(\xi) = (g(b) - g(a)) \cdot f'(\xi)$$

Sign change

$f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in \text{int}D_f$ and $f(a) = 0$, ($K_r(a) \subset D_f, r > 0$)

1. the function f has a $(-, +)$ sign change if

$$f(x) \leq 0 \leq f(t), \quad (x, t \in K_r(a), x < a < t)$$

2. the function f has a $(+, -)$ sign change if
 $f(x) \geq 0 \geq f(t), \quad (x, t \in K_r(a), x < a < t)$

First-order sufficient condition

$f \in \mathbb{R} \rightarrow \mathbb{R}, a \in \text{int}D_f$ and $f \in D\{x\}, (x \in K_r(a) \subset D_f, r > 0)$

1. f' has a $(+, -)$ sign change at $a \Rightarrow f$ has a local maximum at a .
2. f' has a $(-, +)$ sign change at $a \Rightarrow f$ has a local minimum at a .

4.2 Monotonicity

$f \in \mathbb{R} \rightarrow \mathbb{R}$

Monotone increasing

f is monotone increasing ($\nearrow$) if $\forall x, t \in D_f, x < t : f(x) \leq f(t)$.

If $f(x) < f(t)$, then f is strictly monotone increasing ($\uparrow$).

Monotone decreasing

f is monotone decreasing ($\searrow$) if $\forall x, t \in D_f, x < t : f(x) \geq f(t)$.

If $f(x) > f(t)$, then f is strictly monotone decreasing ($\downarrow$).

Relationship between derivative and monotonicity

$I \subset \mathbb{R}$ open interval, $f : I \rightarrow \mathbb{R}, f \in D$

$\Rightarrow$

1. $f \nearrow \Leftrightarrow f' \geq 0$
2. $f \searrow \Leftrightarrow f' \leq 0$
3. f constant $\Leftrightarrow \forall x \in I : f'(x) = 0$
4. $\forall x \in I : f'(x) > 0 \Rightarrow f \uparrow$
5. $\forall x \in I : f'(x) < 0 \Rightarrow f \downarrow$

4.3 Shape relations

Convexity, concavity

$I \subset \mathbb{R}$ interval, $f : I \rightarrow \mathbb{R}$

- f is convex if
 $\forall a, b \in I \quad \forall 0 \leq \lambda \leq 1 : f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b)$
- f is concave if
 $\forall a, b \in I \quad \forall 0 \leq \lambda \leq 1 : f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$

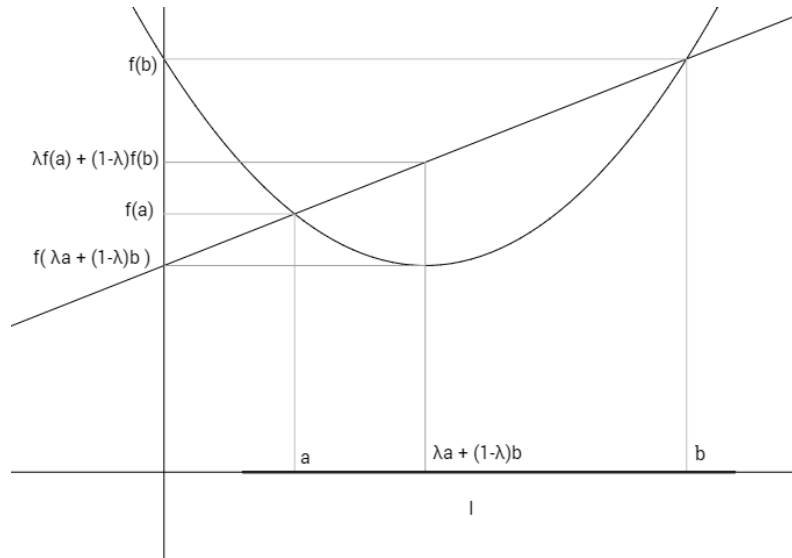


Figure 1: Convex function

Convexity and derivative

$I \subset \mathbb{R}$ open interval, $f : I \rightarrow \mathbb{R}$, $f \in D$

- f convex $\Leftrightarrow f' \nearrow$
- f concave $\Leftrightarrow f' \searrow$

Inflection

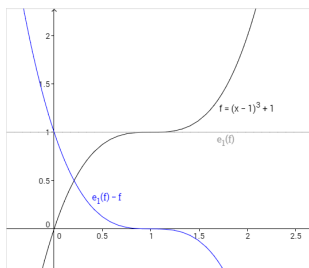
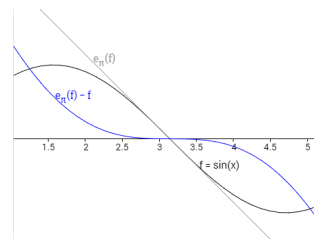
$f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in \text{int}D_f$, $f \in D\{a\}$:

Tangent line at a point

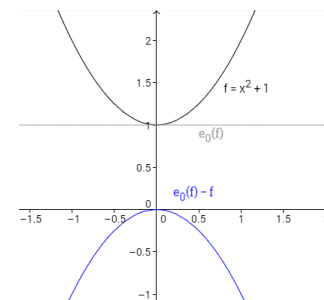
$$e_a(x) := f(a) + f'(a)(x - a) \quad (x \in \mathbb{R})$$

Inflection

f has an inflection at a if $f - e_a(f)$ changes sign at a .

(a) Inflection of the x^3 function

(b) Inflection of the sine function

(c) x^2 has no inflection

4.4 Functions differentiable several times

Second derivative

$f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in \text{int}D_f$, and

$f \in D\{x\} \quad (x \in K_r(a), r > 0)$, and $f' \in D\{a\}$

Then f is twice differentiable at a and $f''(a) := (f')'(a)$ is the second derivative of f .

Higher-order differentiation

Similarly to the previous:

$f \in \mathbb{R} \rightarrow \mathbb{R}$, $a \in \text{int}D_f$, and

$f \in D^n\{x\} \quad (x \in K_r(a), r > 0)$, and $f^{(n)} \in D\{a\}$, $(1 \leq n \in \mathbb{N})$

Then f is $(n+1)$ -times differentiable at a and $f^{(n+1)}(a) := (f^{(n)})'(a)$ is the $(n+1)$ -th derivative of f .

Second-order sufficient condition (for the existence of a local extremum)

for a function $f \in D^2\{a\}$ with $f'(a) = 0$ and $f''(a) \neq 0$

$\Rightarrow f$ has a strict local extremum at a .

If $f''(a) < 0 \Rightarrow$ strict local maximum.

If $f''(a) > 0 \Rightarrow$ strict local minimum.

5 Riemann integral, integration by parts, integration by substitution.**Primitive function**

$I \subset \mathbb{R}$ open interval, $f \in I \rightarrow \mathbb{R}$

If $\exists F : I \rightarrow \mathbb{R}$ such that $F \in D$, $F' = f$

then F is a primitive function of f .

Indefinite integral

Let $\int f := \int f(x)dx := \{F : I \rightarrow \mathbb{R}, F \in D \text{ and } F' = f\}$ be the indefinite integral of f .

Definite integral (Riemann integral)

$-\infty < a < b < \infty$, $f : [a, b] \rightarrow \mathbb{R}$, f bounded

- $\tau \subset [a, b]$ is a partition if τ is finite and $a, b \in \tau$

Then $\tau = \{x_0, x_1, \dots, x_n\} (n \in \mathbb{N})$, where $a := x_0 < x_1 < \dots < x_n := b$

- $m_i := m_i(f) := \inf\{f(x) : x_i \leq x \leq x_{i+1}\} (i = 0..n-1)$, and

$M_i := M_i(f) := \sup\{f(x) : x_i \leq x \leq x_{i+1}\} (i = 0..n-1)$

and:

$$s(f, \tau) := \sum_{i=0}^{n-1} m_i(x_{i+1} - x_i) - \text{lower sum}$$

$$S(f, \tau) := \sum_{i=0}^{n-1} M_i(x_{i+1} - x_i) - \text{upper sum}$$

- $\mathfrak{F} := \{\tau \subset [a, b] \text{ partition}\}$
- The set $\{s(f, \tau) : \tau \in \mathfrak{F}\}$ is bounded above and $\forall \mu \in \mathfrak{F} : S(f, \mu)$ is an upper bound, and the set $\{S(f, \tau) : \tau \in \mathfrak{F}\}$ is bounded below and $\forall \mu \in \mathfrak{F} : s(f, \mu)$ is a lower bound
- So let:
 $I_*(f) := \sup\{s(f, \tau) : \tau \in \mathfrak{F}\}$ - lower Darboux index, and
 $I^*(f) := \inf\{S(f, \tau) : \tau \in \mathfrak{F}\}$ - upper Darboux index

$$\Rightarrow \forall \tau, \mu \in \mathfrak{F} : s(f, \tau) \leq I_*(f) \leq I^*(f) \leq S(f, \mu)$$

The function f is Riemann integrable ($f \in R[a, b]$) if $I_*(f) = I^*(f)$; then let

$\int_a^b f := \int_{[a, b]} f := \int_a^b f(x)dx := I_*(f) = I^*(f)$ be the Riemann integral (definite integral) of the function f .

Integration by parts

- In the indefinite case
 $I \subset \mathbb{R}$ open interval, $f, g : I \rightarrow \mathbb{R}$, $f, g \in D$ and
 fg' has a primitive function (that is, $\int fg' \neq \emptyset$)
 $\Rightarrow \int f'g \neq \emptyset$ and $\int f'g = fg - \int fg'$
- In the definite case
 $f, g \in D[a, b]$
 $f'g, fg' \in R[a, b]$
 $\Rightarrow \int_a^b f'g = f(b)g(b) - f(a)g(a) - \int_a^b fg'$

Integration by substitution

- In the indefinite case
 $I, J \subset \mathbb{R}$ open intervals, $g : I \rightarrow J, g \in D, f : J \rightarrow \mathbb{R}, \int f \neq \emptyset$
 $\Rightarrow \int f \circ g \cdot g' \neq \emptyset$ and $(\int f) \circ g = \int (f \circ g \cdot g')$
- In the definite case
 $f \in C[a, b], g : [\alpha, \beta] \rightarrow [a, b], g \in C^1[\alpha, \beta],$
 $g(\alpha) = a, g(\beta) = b$
 $\Rightarrow \int_a^b f = \int_\alpha^\beta (f \circ g \cdot g')$

6 Newton–Leibniz formula

$f \in R[a, b], \exists F : [a, b] \rightarrow \mathbb{R}, F$ continuous and $F \in D\{x\}, F'(x) = f(x), (a < x < b)$
 $\Rightarrow \int_a^b f = F(b) - F(a)$